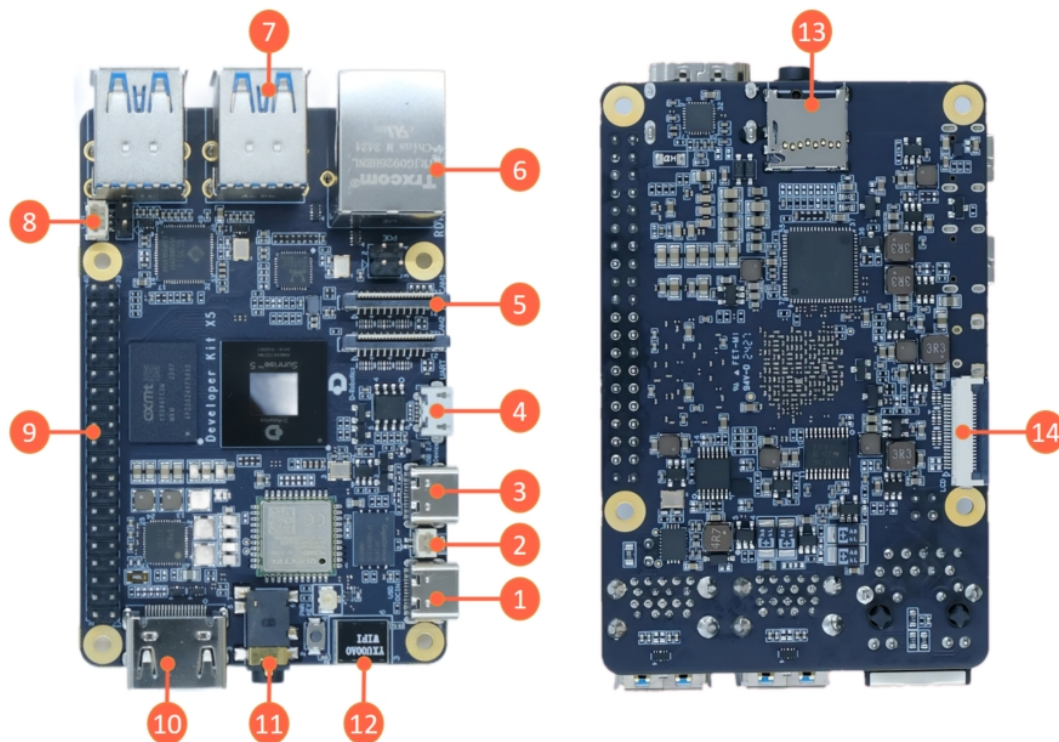


Hardware Interface Description

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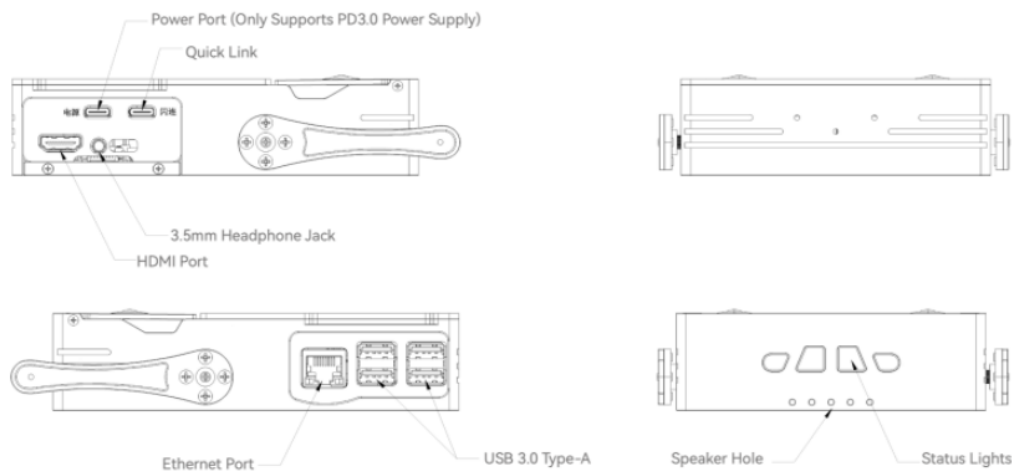
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1. RDK X5 Interface Distribution



No.	Function	No.	Function	No.	Function
1	Power Supply Interface (USB Type C)	2	RTC Battery Interface	3	Easy Connect Interface (USB Type C)
4	Debug Serial Port (Micro USB)	5	2-way MIPI Camera Interface	6	Gigabit Ethernet Port, supports PoE
7	4-way USB 3.0 Type A Interface	8	CAN FD High-speed Interface	9	40PIN Interface
10	HDMI Display Interface	11	Multi-standard Compatible Headphone Jack	12	Onboard Wi-Fi Antenna
13	TF Card Interface (Bottom)	14	LCD Display Interface (MIPI DSI)		

2. RDK MagicBox Interface, Buttons and Indicator Lights Description



3. Power Interface

The development board provides one USB Type C interface (interface 1) as the power supply interface. It is necessary to use a power adapter supporting **5V/5A** to power the development board. After connecting the power adapter to the development board, **the green power indicator and orange indicator on the development board light up**, indicating the development board's power supply is normal.

Warning

Please do not use the computer USB interface to power the development board, otherwise it may cause abnormal power-off and repeated restarts due to insufficient power supply.

4. Debug Serial Port (Flash Connect Port)

The development board provides one debug serial port to implement serial login and debugging functions. The parameter configuration of the computer serial tool is as follows:

The development board provides one debug serial port (interface 4) to implement serial login and debugging functions. The parameter configuration of the computer serial tool is as follows:

- Baud rate: 115200
- Data bits: 8
- Parity: None
- Stop bits: 1
- Flow Control: None

When connecting the serial port, you need to connect the development board interface 4 to the PC via a Micro-USB cable.

Usually, users need to install the CH340 driver on the computer when using this interface for the first time. Users can search for `CH340 serial driver` keyword to download and install.

5. Wired Ethernet Port

The development board provides one Gigabit Ethernet interface (interface 6), supporting 1000BASE-T and 100BASE-T standards, defaulting to static IP mode with IP address `192.168.127.10`. To confirm the development board's IP address, you can log in to the device via serial port and use the `ifconfig` command to check the `eth0` network interface configuration.

Additionally, this interface supports PoE (Power over Ethernet) function, which can transmit data and power simultaneously through the network cable without additional power cables, making device installation more convenient and flexible.

6. HDMI Display Interface

The development board provides one HDMI (interface 10) display interface, supporting up to 1080P resolution. The development board outputs Ubuntu system desktop through the HDMI interface on the monitor (Ubuntu Server version displays logo icon). Additionally, the HDMI interface supports real-time camera and network stream display functions.

7. USB Interface

The development board implements multi-way USB interface expansion through hardware circuits, meeting users' needs for multi-way USB device access. The interface descriptions are as follows:

Interface Type	Interface No.	Interface Quantity	Interface Description
USB 3.0 Type C	Flash Connect Port	1-way	USB Device mode, used to connect to host for ADB, Fastboot, system flashing and other functions
USB 3.0 Type A	Interface 7	4-way	USB Host mode, used to connect USB 3.0 peripherals

Connecting USB Flash Drive

The development board USB Type A interface (interface 7) supports USB flash drive function, can automatically detect USB flash drive connection and mount it. The default mount directory is `/media/sda1`.

Connecting USB Serial Adapter Board

The development board USB Type A interface (interface 7) supports USB serial adapter board function, can automatically detect USB serial adapter board connection and create device nodes `/dev/ttyUSB*` or `/dev/ttyACM*` (asterisk represents numbers starting from 0).

8. MIPI CSI Interface

The development board provides 2-way MIPI CSI interfaces (interface 5), which can realize 2-way MIPI camera access, supporting dual-camera access. Currently, the development board is adapted to various specifications of camera modules. The module models and specifications are as follows:

No.	Sensor	Resolution	FOV	I2C Device Address
1	IMX219	800W		
2	OV5647	500W		

The camera module connects to the development board via FPC ribbon cable. Note that the blue side of the cable ends should be facing up and inserted into the connector.

After installation is complete, users can use the `i2cdetect` command to confirm whether the module I2C address can be detected normally.

Important Note: 1. It is strictly prohibited to plug or unplug the camera when the development board is not powered off, otherwise the camera module can be easily burned.

2. The 2 MIPI CSI interfaces of this product have been connected with a dual-camera. The dual-camera is used for image perception and depth information acquisition, supporting gesture recognition, target positioning, spatial measurement and other usage scenarios.

9. 3.5mm Headphone Jack

This headphone jack, as shown in the RDK MagicBox interface diagram, is used for audio input and output.

10. Speaker Vents

This speaker vent, as shown in the RDK MagicBox interface diagram, is used for audio output.

11. Microphone

The RDK MagicBox system has a built-in microphone used to receive voice commands when running the official voice interaction demo.

12. Function Buttons and Status Lights

The RDK MagicBox system has function buttons and status lights. The status lights are shown in the figure. The buttons are located beside the metal shell above the fan.

Power on:

White lights light up in sequence, wait approximately 50s

White light flashes with sound effect: Startup complete

Function buttons: Start preset examples; with the light strip facing the user from left to right:

Button 1: Start dual-camera depth estimation function, light turns red

Button 2: Start gesture interaction function, light turns green

Button 3: Start voice interaction function, light turns blue

13. LCD Display Interface

RDK X5 provides one MIPI DSI LCD display interface (interface 14), which can be used for LCD screen and other access. The interface uses a 15-pin FPC connector, which can directly connect to various Raspberry Pi LCD screens.

14. Micro SD Interface

The development board provides 1-way Micro SD storage card interface. It is recommended to use a storage card with at least 32GB capacity to meet the installation requirements of the Ubuntu operating system and related function packages. The default package for this product is a 32GB memory card package, which meets the operation of our cases.

Warning

Hot-swapping of the TF storage card is prohibited during development board use, otherwise it may cause abnormal system operation or even damage to the storage card file system.

15. Wi-Fi Antenna Interface

The development board's wireless network supports two configurations: onboard and external antennas. Normally, **the onboard antenna can meet usage needs**. When the development board is installed with a metal shell, it is necessary to connect an external antenna to the antenna interface beside (interface 12) to enhance signal strength.

16. CANFD Interface

RDK X5 development board provides CANFD interface, which can be used for CAN and CAN FD communication.

17. 40PIN Header Interface

RDK X5 development board provides a 40PIN header interface, with I/O signal using 3.3V level design. The pin definitions are compatible with Raspberry Pi and other products. The detailed pin definitions and multiplexing relationships are as follows:

RDX X5 Board 40Pin Table																	
Reuse function3	Reuse function2	Reuse function1	Reuse function 0	Function Description	XS Pin Number	BCM Encoding	CVM Function	Physical Pin Board Encoding	CVM Function	BCM Encoding	XS Pin Number	Function Description	Reuse function 0	Reuse function1	Reuse function2	Reuse function3	
				3.3V power signal			VDD JV3	1 2	VDD 5V			5V power signal					
	LSIO_GPIO_11	SWA5	UART3_TXD	I2CD data signal	387	2	I2C5_SDA	3 4	VDD 5V			5V power signal					
	LSIO_GPIO_10	SC15	UART3_RXD	I2CD clock signal	389	3	I2C5_SCL	5 6	GND			GND signal					
		DSP_GPIO_39	DSP_MCLK1	I2SD MCLK1 clock signal	420	4	I2S1_MCLK1	7 8	UART1_TXD	14	383	UART1 send signal	UART1_TXD		LSIO_GPIO_5		
				GND signal			GND	9 10	UART_RXD	15	384	UART1 receive signal	UART1_RXD		LSIO_GPIO_4		
	LSIO_GPIO0_1		UART7_TXD	GPIO17 signal	380	17	GPIO17	11 12	I2S1_BCLK	18	421	I2S1 BCLK clock signal	I2S1_BCLK	DSP_GPIO_10			
	LSIO_GPIO0_0		UART7_RXD	GPIO22 signal	379	27	GPIO27	13 14	GND			GND signal					
	LSIO_GPIO0_9		UART2_TXD	GPIO22 signal	388	22	GPIO22	15 16	GPIO23	23	382	GPIO23 signal	UART0_RTS	UART6_TXD	LSIO_GPIO_3		
				3.3V power signal			VDD JV3	17 18	GPIO24	24	402	GPIO24 signal	SPI2_MOSI	LSIO_GPIO_23		LSIO_PWM_OUT3	
	J16_TDO	LSIO_GPIO0_19	SPI1_MOSI	SPI1 MOSI signal	398	10	SPI1_MOSI	19 20	GND			GND signal					
	J16_TDI	LSIO_GPIO0_18	SPI1_MISO	SPI1 MISO signal	397	9	SPI1_MISO	21 22	GPIO25	25	387	GPIO25 signal	UART2_RXD		LSIO_GPIO_8		
	J16_TCK	LSIO_GPIO0_16	SPI1_SCLK	SPI1 CLK signal	395	11	SPI1_SCLK	23 24	SPI_CSNO	8	394	SPI1 SSN1 signal	SPI1_CSN1	LSIO_GPIO0_15	J16_TMS		
				GND signal			GND	25 26	SPI_CSNO	7	396	SPI1 SSN0 signal	SPI1_CSN0	LSIO_GPIO0_17	J16_TRSTN		
LSIO_PWM_OUT5		LSIO_GPIO1_8	SSNO	I2C3 clock signal	355	0	I2C0_SDA	27 28	I2C0_SCL	1	354	I2C0 signal	SCL0	LSIO_GPIO1_7		LSIO_PWM_OUT4	
LSIO_PWM_OUT0		LSIO_GPIO0_20	SPI2_SCLK	GPIO5 signal	399	5	GPIO5	29 30	GND			GND signal					
LSIO_PWM_OUT1	TIME_SYNC2	LSIO_GPIO0_21	SPI2_SSN	GPIO6 signal	400	6	GPIO6	31 32	PWM6	12	356	PWM6 signal	SCL1	LSIO_GPIO1_9	TIME_SYNC1	LSIO_PWM_OUT6	
LSIO_PWM_OUT7		LSIO_GPIO1_10	SDA1	PWM0 signal	357	13	PWM7	33 34	GND			GND signal					
		DSP_GPIO_11	I2S1_LRCK	I2SD LRCK signal	422	19	I2S1_LRCK	35 36	GPIO16	16	381	UART0_CTS	UART5_RXD		LSIO_GPIO0_2		
		DSP_GPIO_22	SPI2_MISO	GPIO26 signal	401	26	GPIO26	37 38	I2S1_SDIN	20	423	I2S1 DI signal	I2S1_DIN	DSP_GPIO_12			
LSIO_PWM_OUT2				GND signal			GND	39 40	I2S1_SDOUT	21	424	I2S1 DO signal	I2S1_DOUT	DSP_GPIO_13			

The development board 40PIN interface location provides silkscreen netlist for users to compare operations. The PIN1 and PIN40 positions are as follows:

